



IK2221: Project Introduction

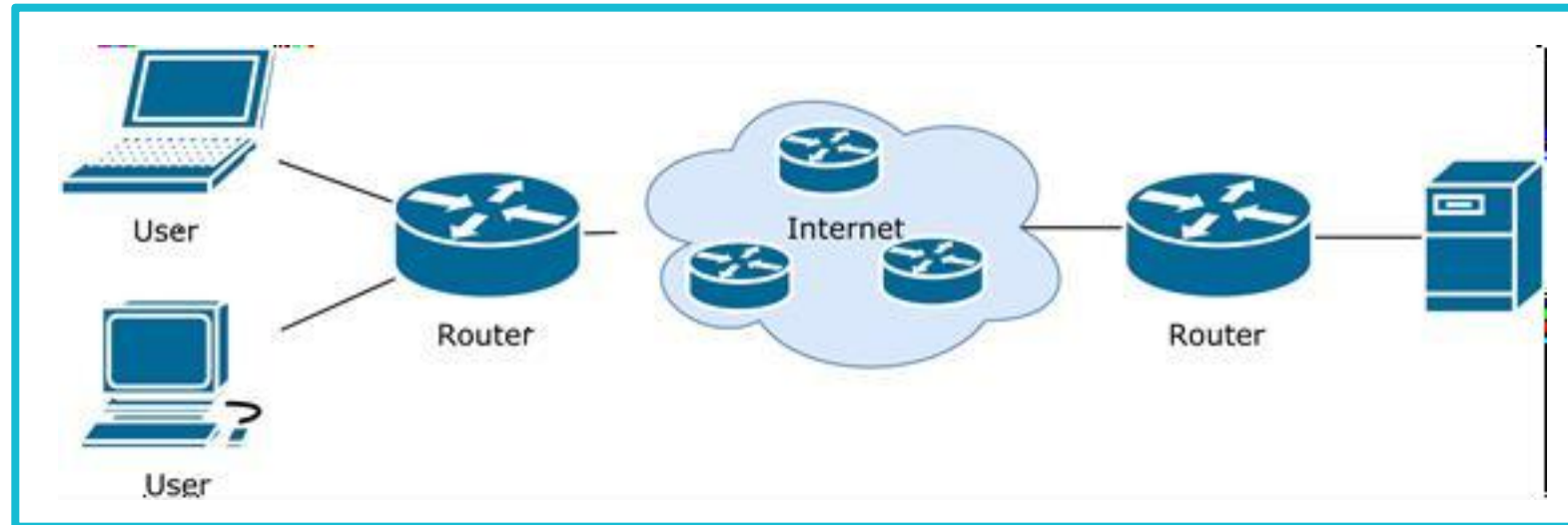
Giacomo Verardo



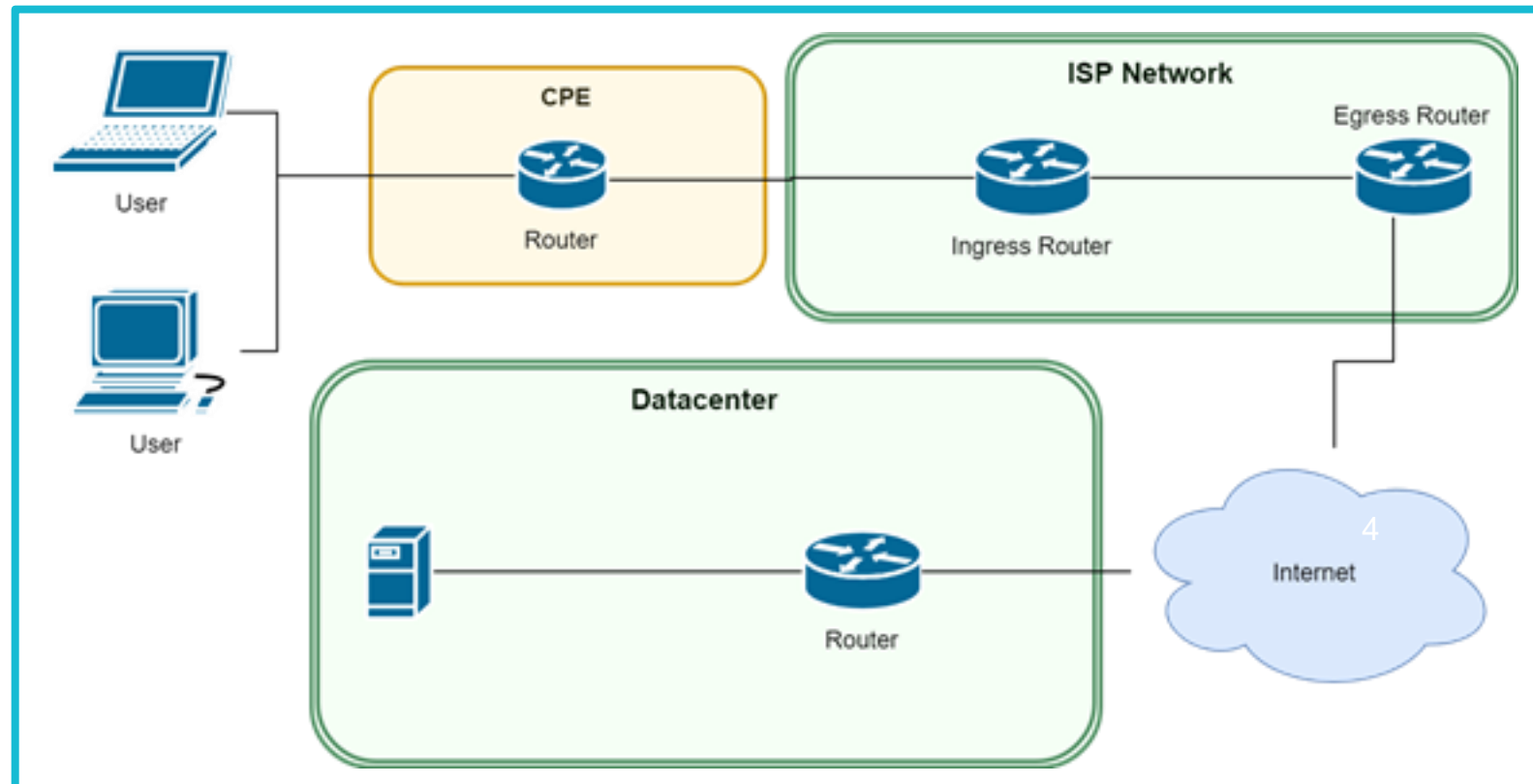
Outline

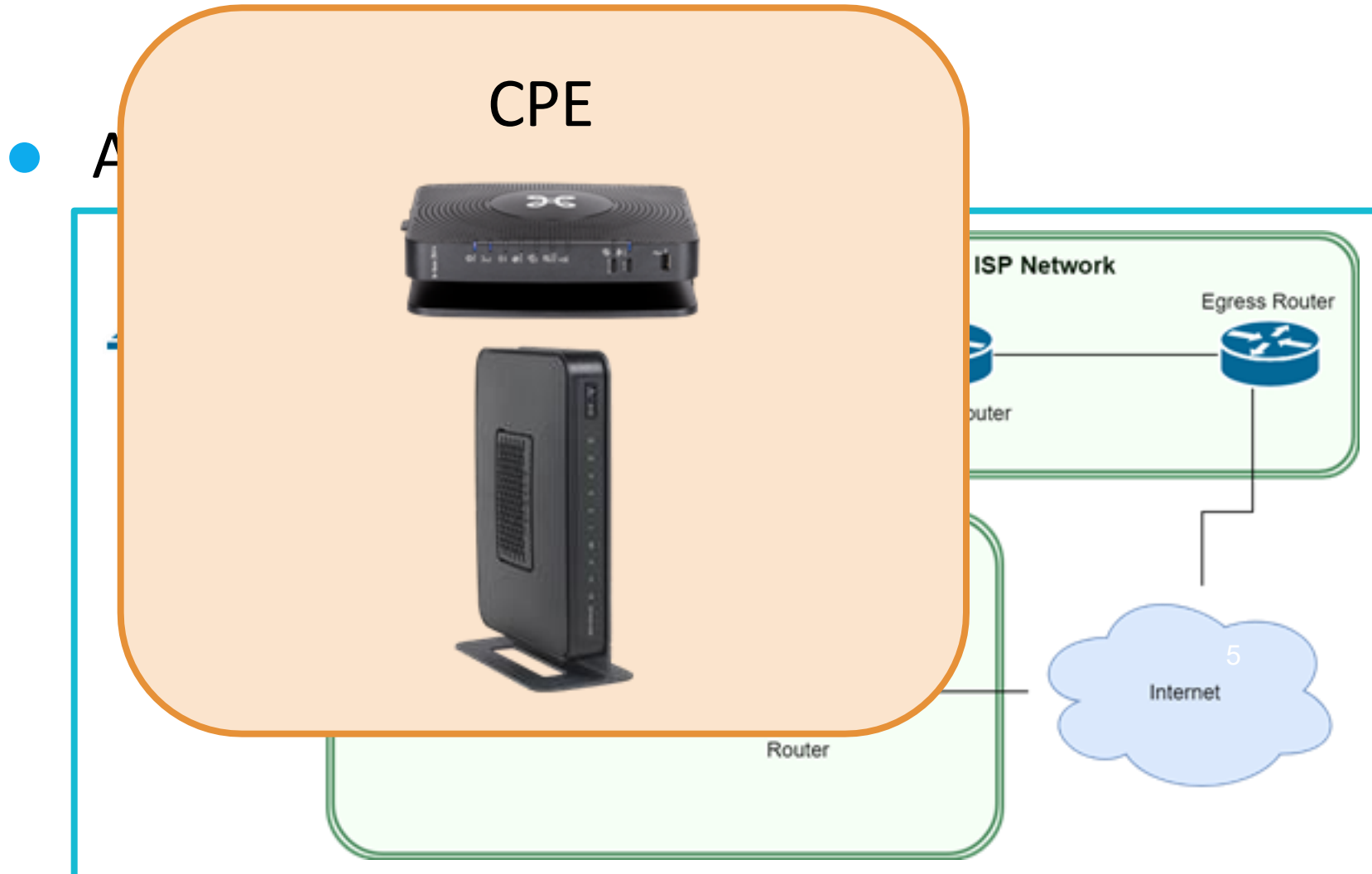
- Network function virtualization (NFV)
- Project description:
 - Phase 1 (NFV)
 - Phase 2 (LLM)
- Practical information
 - Form Groups!
 - Tutorials
- Q&A

- Computers interconnected by router and switches



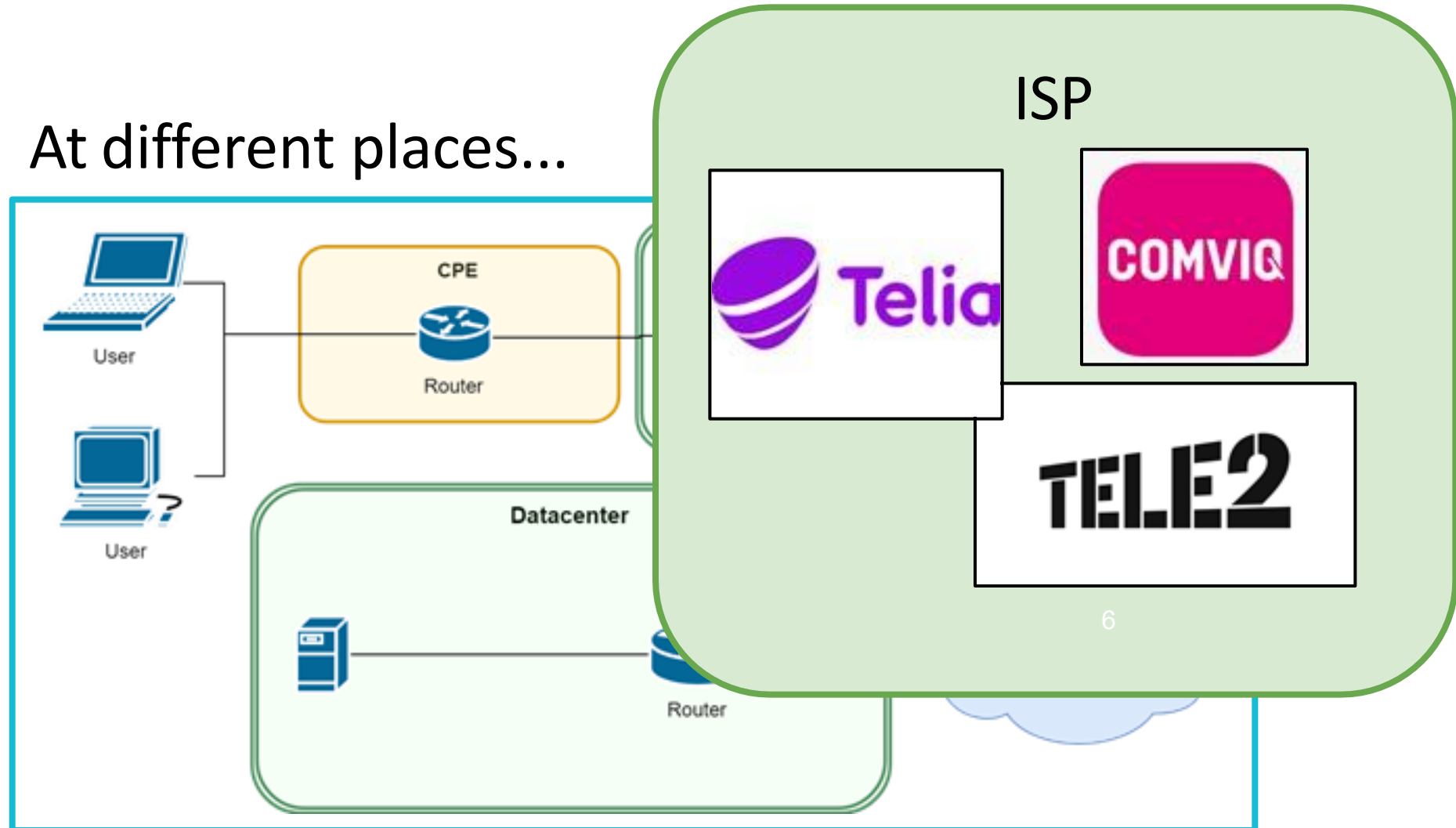
- At different places...



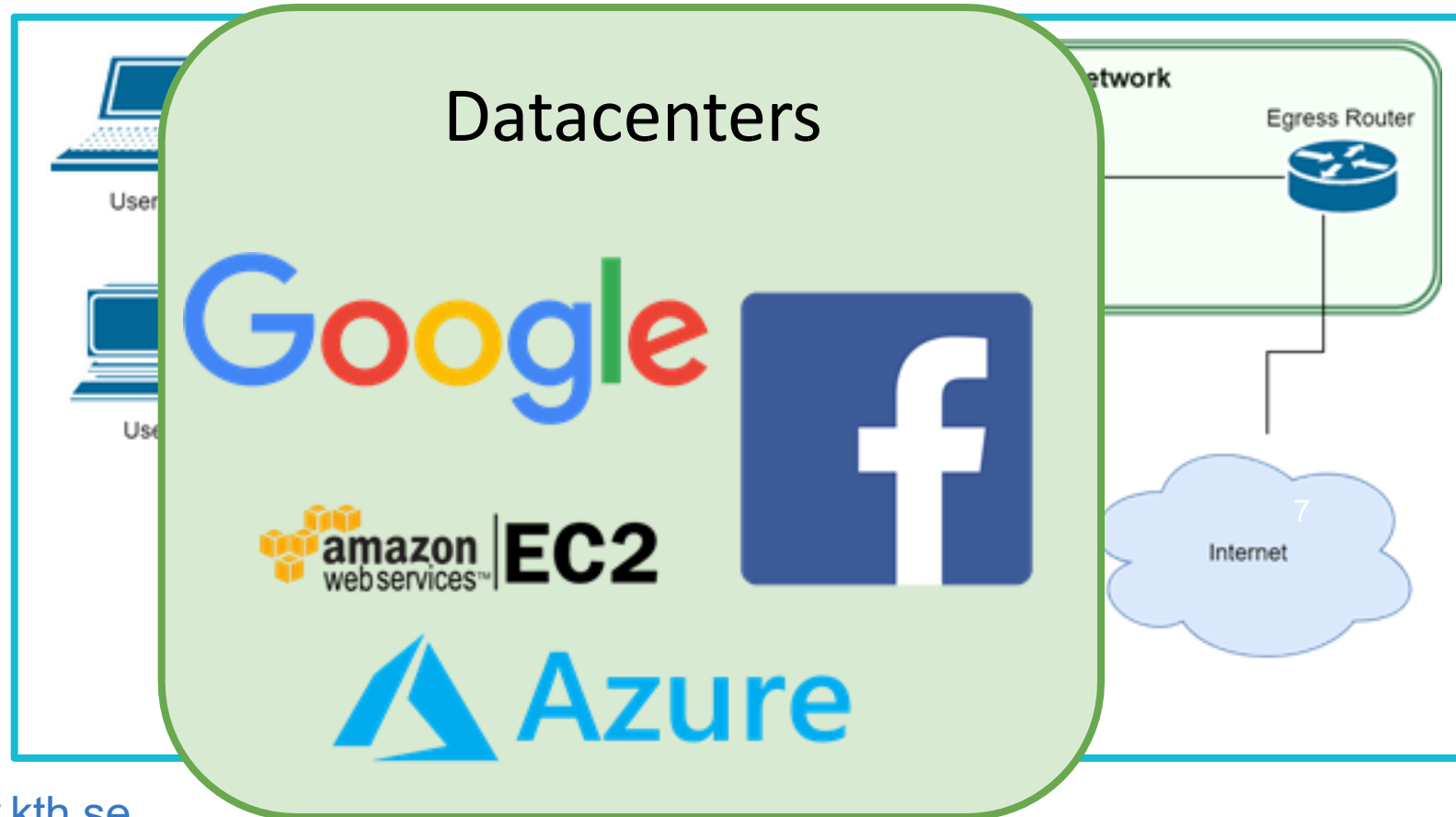


The Internet

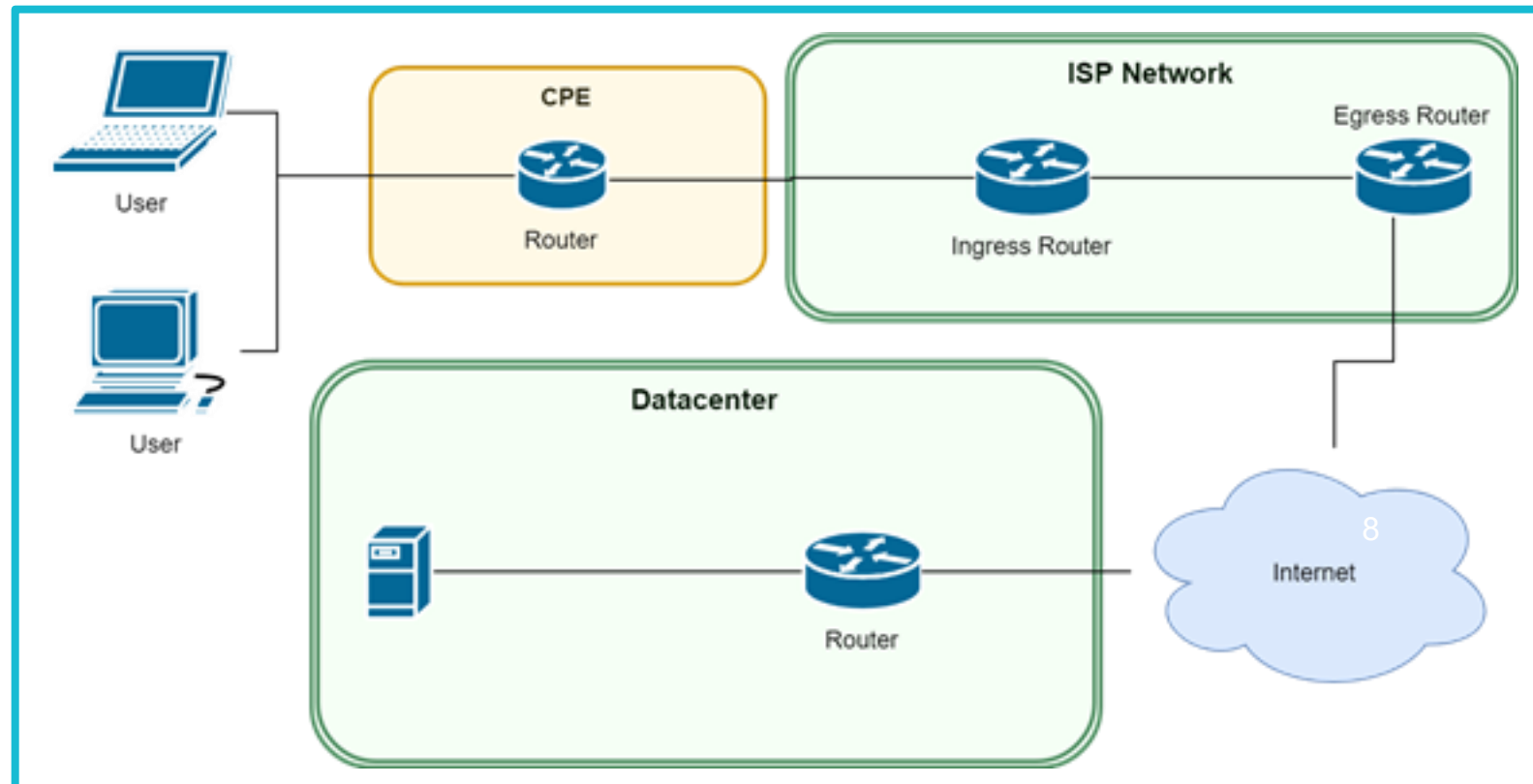
- At different places...



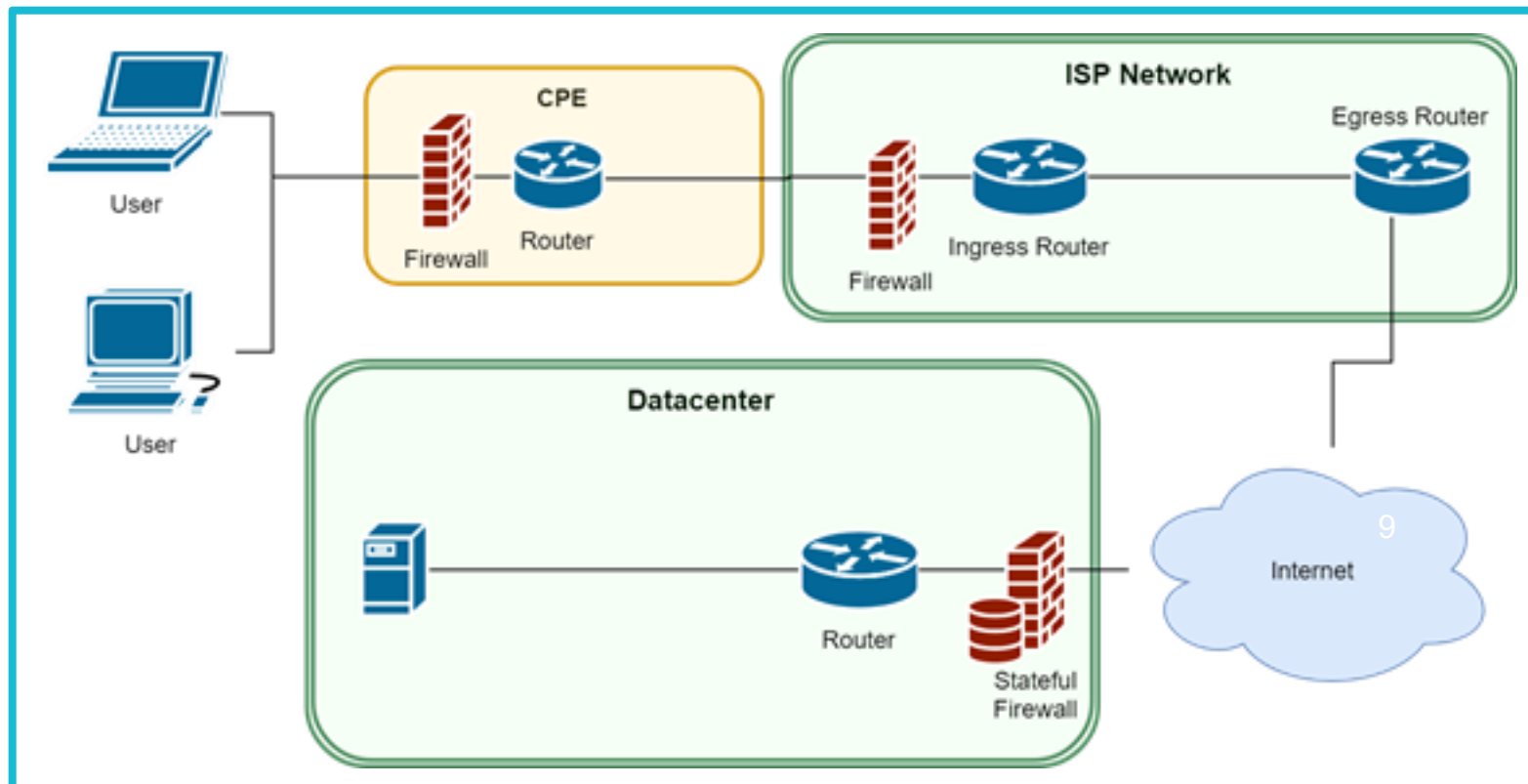
- At different places...



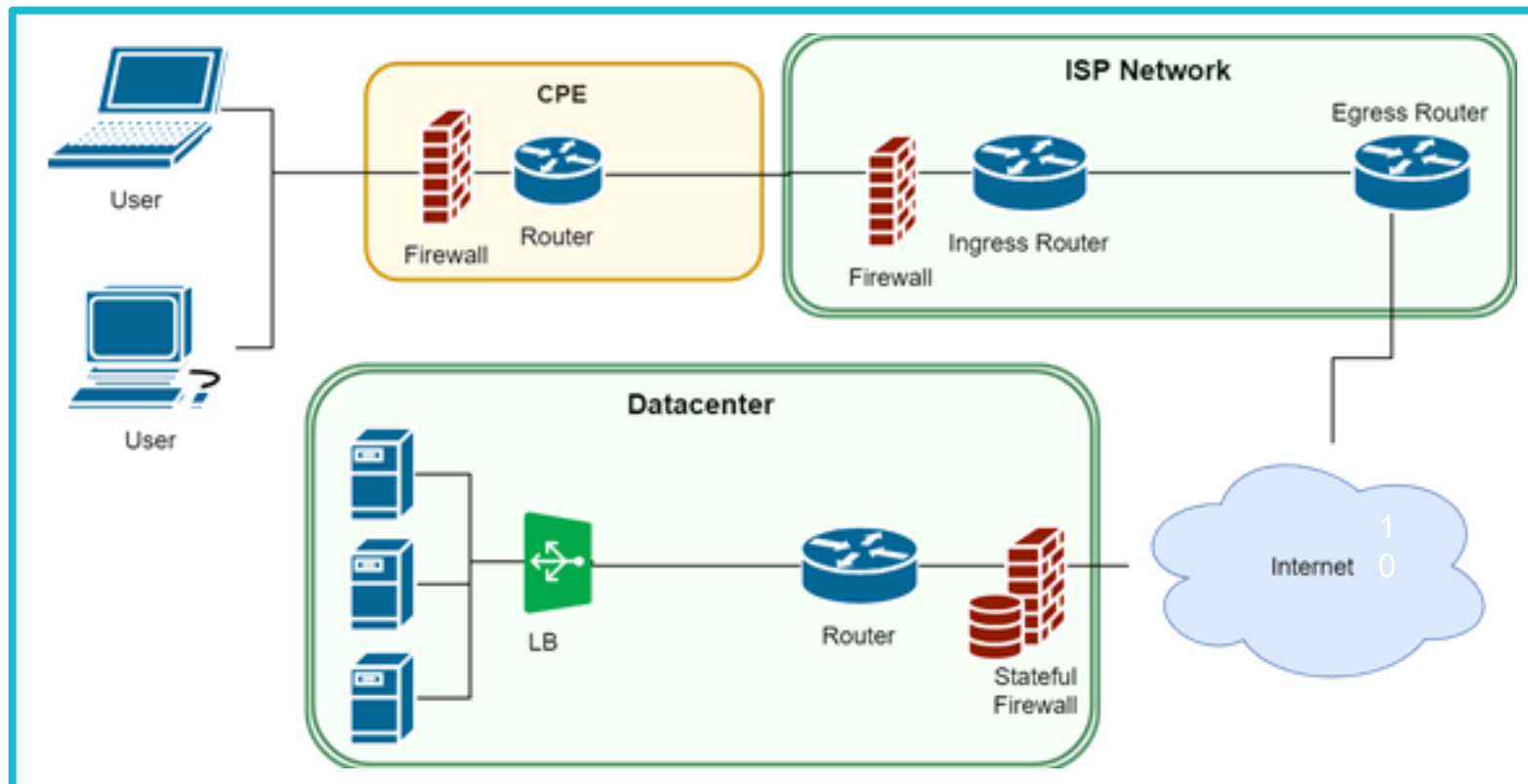
- Only router and switches?



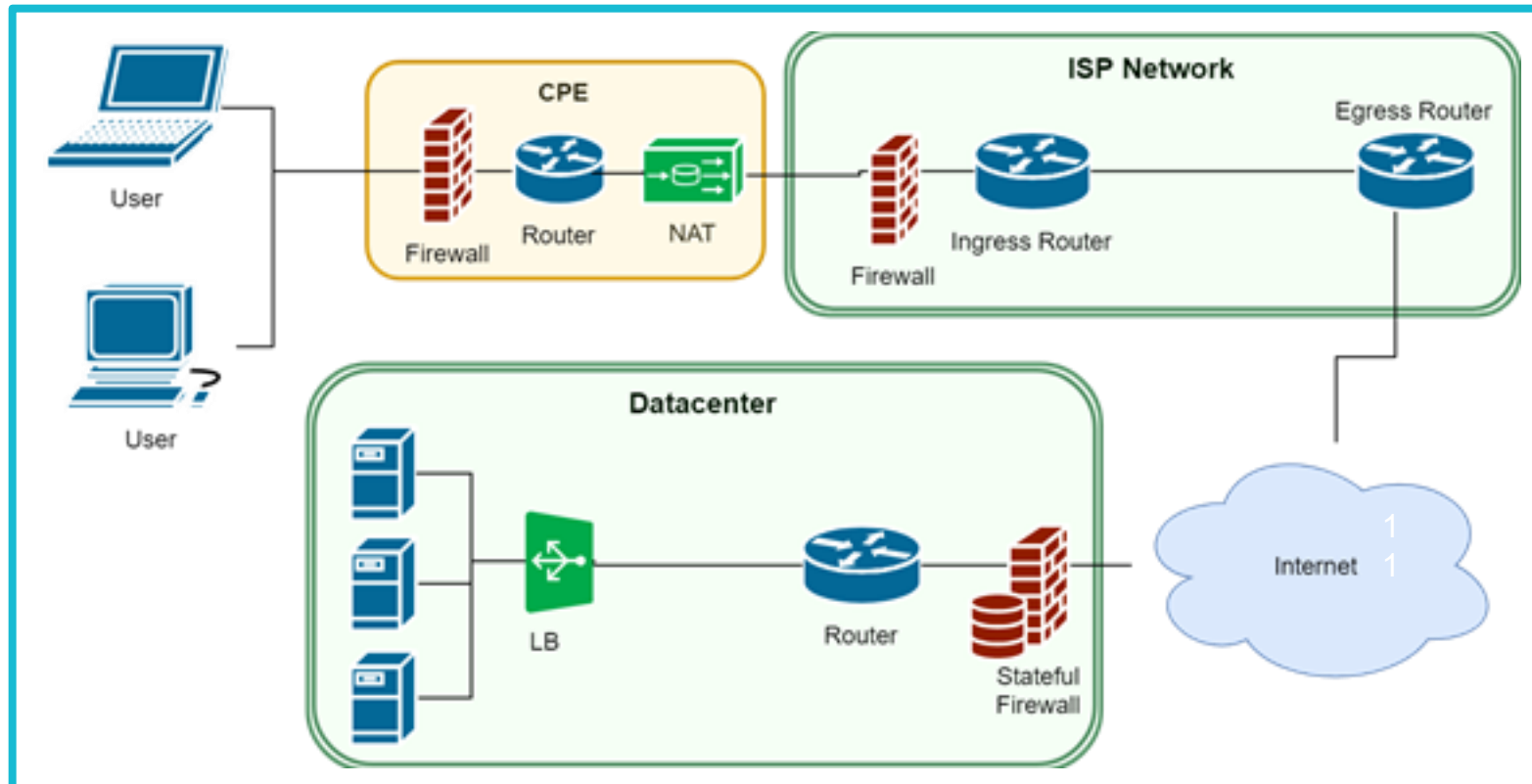
- Also firewalls...



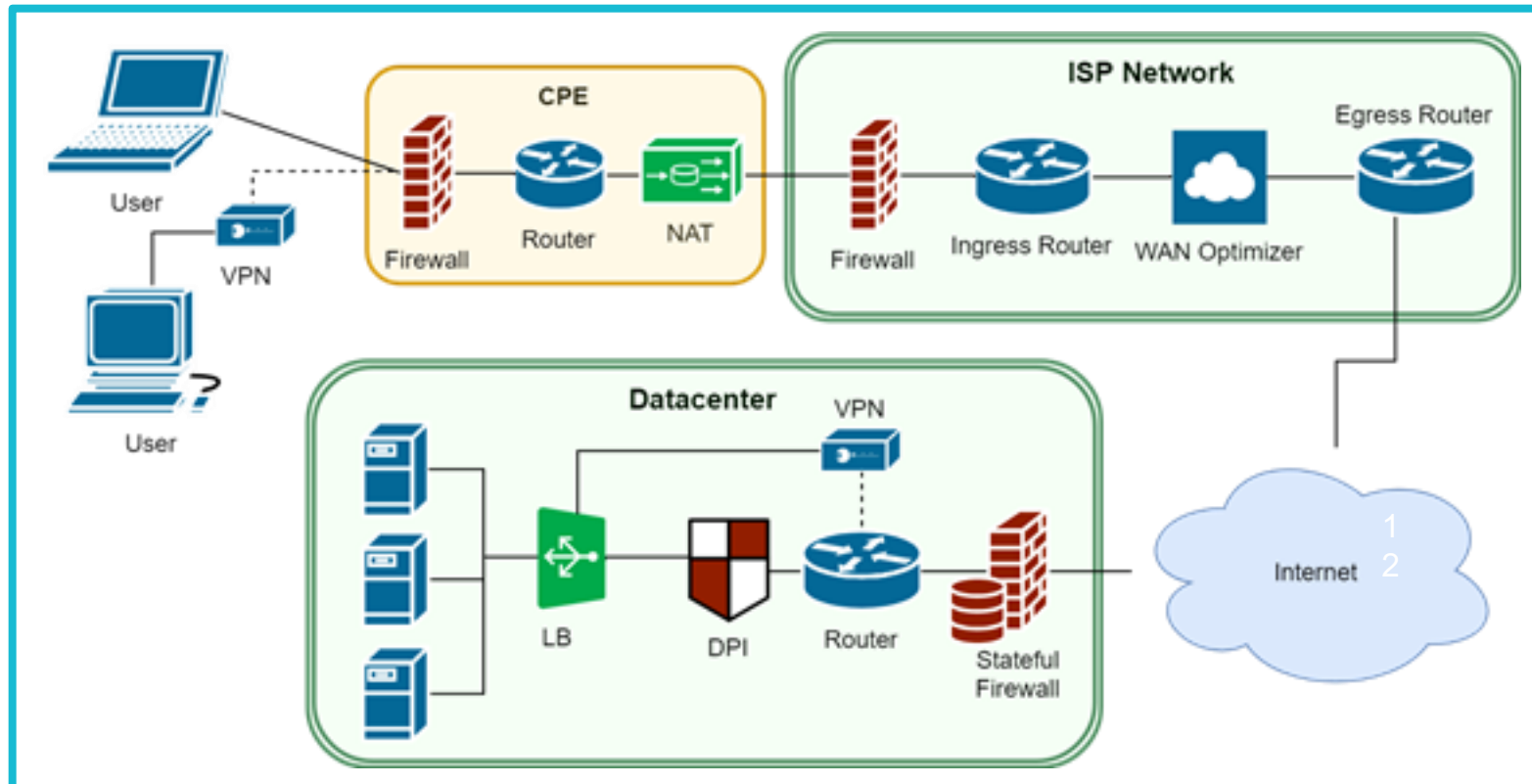
- Also load balancers...



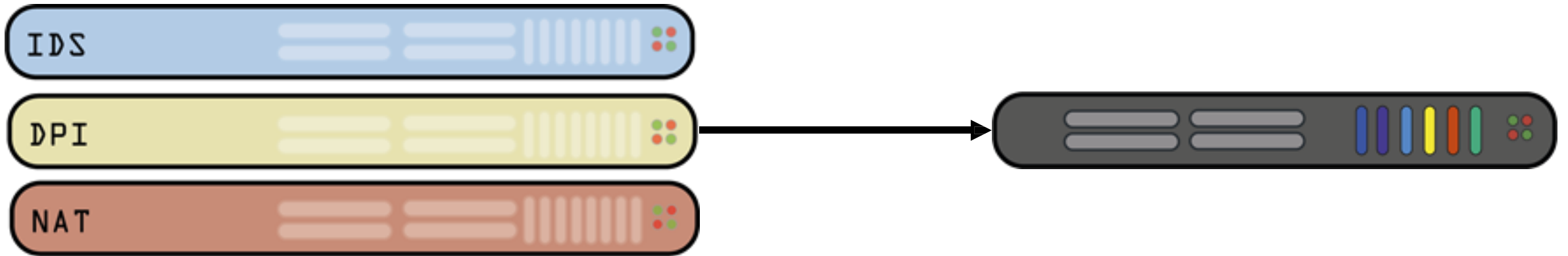
- Also NATs...



- [Sherry12] $\frac{1}{3}$ of network equipments are MiddleBoxes !



Network Function Virtualization

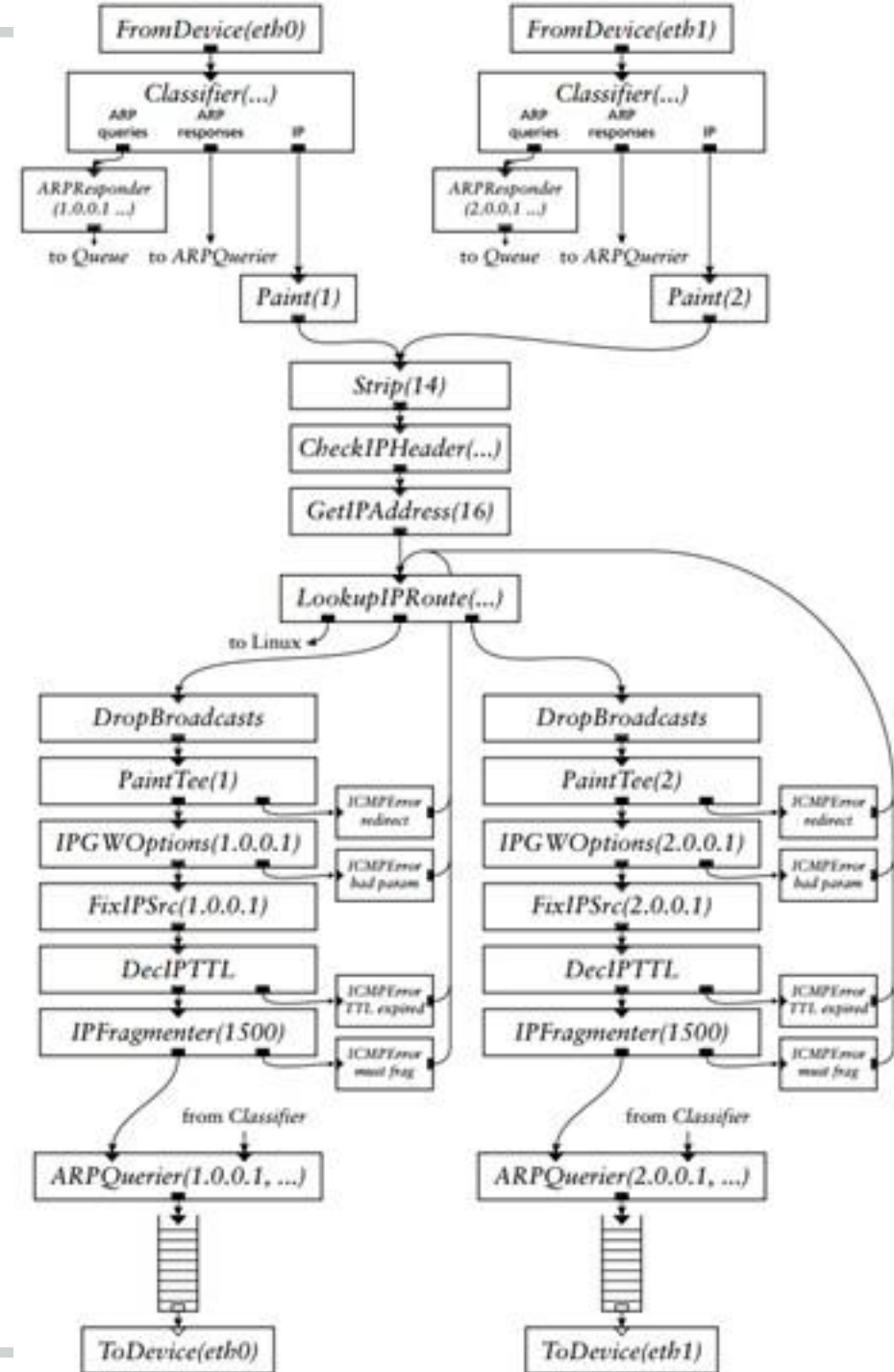


Specialized, inflexible,
expensive boxes

General purpose software

The Click Modular Router

- NFV router only at first
- Many more functions over the last decades





Assignment - Phase 1

- Become familiar with NFV
 - Implement basic network functions.
- Learn:
 - About Click.
 - How the OS networking stack works.
- Deadline: 29th April 17:00 (defense on 30th)

Assignment - Phase 2

- Become familiar with Large Language Model Inferencing
 - Use basic inferencing tools
 - Improve inferencing pipelines with scheduling
- Learn:
 - About the KV cache of LLMs.
 - How inferencing works for LLMs.
- Deadline: End of May (see Canvas for details)

Assignment - Phase 1

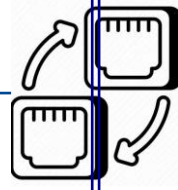
User Zone

h1: 10.0.0.50/24



h2: 10.0.0.51/24

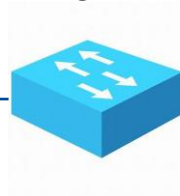
s1



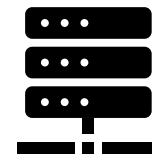
napt

10.0.0.1/24 $\leftrightarrow$ 100.0.0.1/24

s2



ids

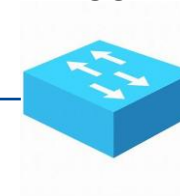


Insp:
100.0.0.30/24

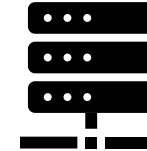


lb1
HTTP service@
100.0.0.45:80

s3



llm1



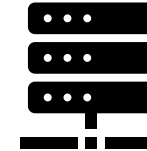
100.0.0.40
/24

llm2



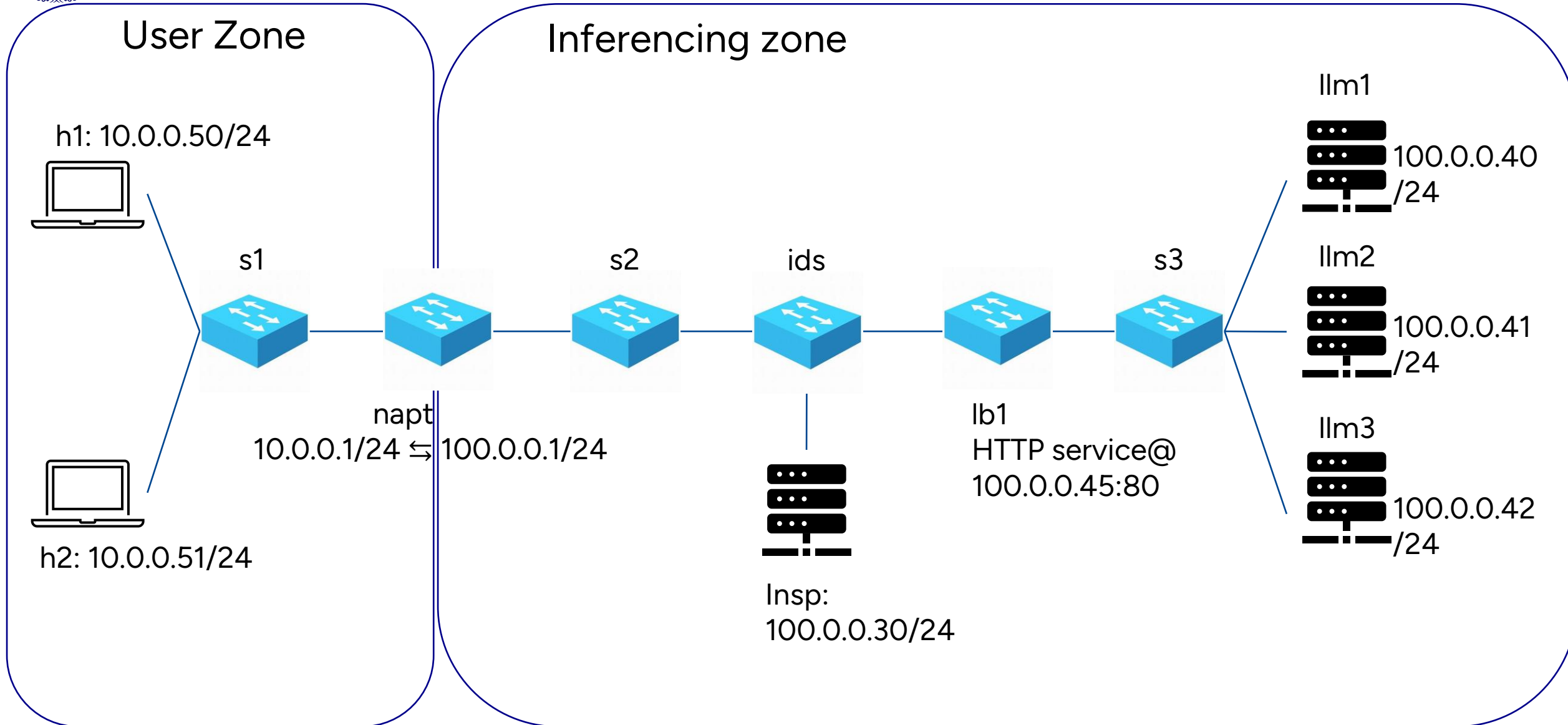
100.0.0.41
/24

llm3



100.0.0.42
/24

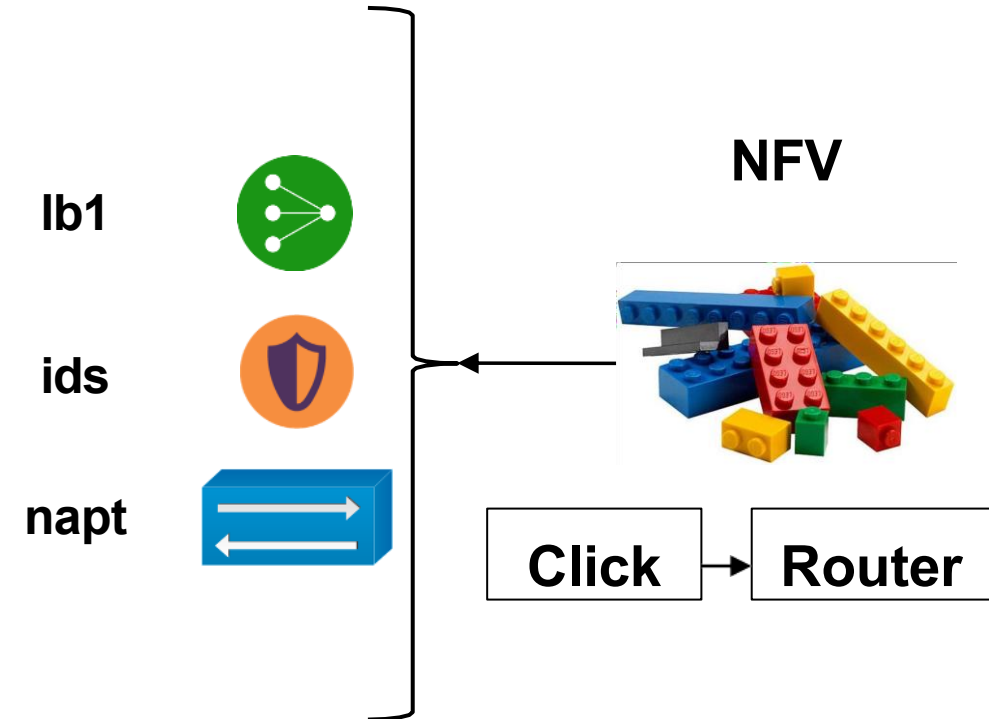
Assignment - Phase 1



Assignment - Phase 1

- Implement these NFs
 - Using Click

Read the detail in the
project description





Get ready !

- Form groups of 4 students (1 if you are a PhD student)
 - Register in Canvas.
- Download the course VM and test if you have any issue
 - Link is in Canvas.
 - Fill Python/Linux/Networking gaps (if any)
- If you have any problem related to the project:
 - Please ask in the discussion forum (other students can benefit from your doubts as well)
 - Ask during exercise sessions



If you have not done any programming

- Python Tutorial by Giacomo Verardo (IK2217)
 - [Colab link](#)
- Makefile Tutorial
 - <https://makefiletutorial.com/>
- Git Tutorial by Alireza Farshin (IK2200)
 - [Git.pdf: IK2221/FIK3221 VT25 Networked Systems for Machine Learning](#)
- Previous years recording on Canvas
- Mininet walkthrough: <http://mininet.org/walkthrough/>



Next exercise session

- Click tutorial
- Running the skeleton code on the VM

Thank you!

